SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- CONSTRAINT-BASED PROBLEM SOLVING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO5	Assessment:	ASSESSMENT TOOL1- Constraint-Based Problem Solving
Weightage:	40%	Total Marks:	25
CO5 Statement:	*Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.*		
Student Name:	Moval Raj. B	Reg.No.:	192511176
Department:	B.E-CSE	Date:	

55%

ANNEXURE A

Constraint-Based Problem Solving

1.Construct an I/O communication mechanism for a real-time embedded system with the following constraints:

- CPU utilization must be below 15%
- Multiple sensors generate interrupts every 1 ms
- Use vectored interrupt handling
- Select between memory-mapped and I/O-mapped I/O for optimal latency

2.Design an I/O subsystem under these constraints:

- High-speed storage requires throughput > 3 GB/s
- Must use NVMe over PCIe
- CPU should not handle bulk data transfer
- Integrate DMA controller with interrupt notification

3.Construct a multi-device communication architecture with constraints:

- 6 peripherals using USB, 2 high-speed devices using Thunderbolt
- Avoid interrupt starvation
- Use hardware and software interrupts
- Ensure fair bus arbitration

4.Design a data acquisition system with constraints:

- Continuous data stream at 500 MB/s
- Minimal CPU intervention required
- Must compare DMA transfer vs interrupt-driven I/O
- Use memory-mapped I/O for device registers

5.Construct an interrupt handling mechanism with constraints:

- Support nested interrupts
- Maximum interrupt latency < 10 µs

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5.Construct an interrupt handling mechanism with constraints:

- Support nested interrupts
- Maximum interrupt latency < 10 μs
- Use vectored interrupts for high-priority devices
- Use software interrupts for background tasks

Answers

PROBLEM 1: REAL-TIME I/O COMMUNICATION MECHANISM

Aim

To design an efficient I/O communication mechanism for a real-time embedded system under strict CPU and latency constraints.

Constraints

- CPU utilization < 15%
- Sensors generate interrupts every 1 ms
- Use vectored interrupt handling
- Choose between memory-mapped and I/O-mapped I/O

Design Solution

Architecture

- Use **Vectored Interrupt Controller (VIC)**
- Assign **dedicated interrupt vectors** to each sensor
- Use **Memory-Mapped I/O (MMIO)** for faster access

Working Mechanism

1. Sensors generate interrupts every 1 ms
2. Interrupt controller directly jumps to ISR using vector table
3. ISR processes minimal data and offloads tasks
4. Use buffering to reduce CPU load

Diagram

Sensors → Interrupt Controller → CPU (ISR)

↓

Memory Buffer

Justification

- Vectored interrupts reduce latency
- MMIO provides faster access than I/O-mapped I/O
- CPU load reduced by short ISR execution

Conclusion

System achieves **low latency and CPU utilization below 15%**, making it suitable for real-time applications.

PROBLEM 2: HIGH-SPEED I/O SUBSYSTEM DESIGN

Aim

To design a high-performance storage subsystem with throughput > 3 GB/s.

Constraints

- Throughput > 3 GB/s
- NVMe over PCIe required
- CPU should not handle bulk transfer
- Use DMA with interrupt notification

Design Solution

Architecture

- NVMe SSD connected via **PCIe Gen3/Gen4**
- **DMA Controller** handles data transfer
- Interrupt signals completion to CPU

Working Mechanism

1. CPU initiates DMA transfer
2. DMA moves data directly between storage and memory
3. CPU is free during transfer
4. Interrupt generated on completion

Diagram

NVMe SSD → PCIe → DMA Controller → RAM

↓

Interrupt → CPU

Justification

- PCIe provides high bandwidth
- DMA eliminates CPU bottleneck
- Interrupt ensures efficient notification

Conclusion

Achieves **high throughput with minimal CPU usage**, meeting system requirements.

PROBLEM 3: MULTI-DEVICE COMMUNICATION ARCHITECTURE

Aim

To design a communication system supporting multiple devices without starvation.

Constraints

- 6 USB devices
- 2 Thunderbolt devices
- Avoid interrupt starvation
- Use hardware & software interrupts
- Ensure fair bus arbitration

Design Solution

Architecture

- USB Hub + Thunderbolt Controller
- Priority-based interrupt controller
- Round-robin bus arbitration

Working Mechanism

1. Devices send interrupt requests
2. Hardware interrupts handle critical devices
3. Software interrupts handle low-priority tasks
4. Arbiter allocates bus access fairly

Diagram

USB Devices → USB Hub → Controller → CPU

Thunderbolt → TB Controller → CPU

↓

Interrupt Controller + Arbiter

Justification

- Prevents starvation using fair scheduling
- Separates critical and non-critical tasks
- Ensures balanced system performance

Conclusion

System ensures **fair communication and avoids bottlenecks**.

PROBLEM 4: DATA ACQUISITION SYSTEM DESIGN

Aim

To design a high-speed data acquisition system with minimal CPU usage.

Constraints

- Data stream: 500 MB/s
- Minimal CPU intervention
- Compare DMA vs interrupt-driven I/O
- Use memory-mapped I/O

Design Solution

Approach

- Use **DMA-based data transfer**
- MMIO for device register access

Comparison

Feature	DMA Transfer	Interrupt-Driven I/O
CPU Usage	Low	High
Speed	High	Moderate
Efficiency	High	Low
Suitability	Large data	Small data

Diagram

Sensor → ADC → DMA → Memory

↓

CPU

Justification

- DMA handles continuous high-speed data
- MMIO ensures quick register access

Conclusion

DMA is the **optimal solution for high-speed data acquisition systems**.

PROBLEM 5: INTERRUPT HANDLING MECHANISM

Aim

To design an interrupt system with low latency and nested handling.

Constraints

- Support nested interrupts
- Latency < 10 μs
- Use vectored interrupts for high priority
- Use software interrupts for background tasks

Design Solution

Architecture

- Priority-based interrupt controller
- Nested interrupt support
- Separate hardware & software interrupts

Working Mechanism

1. High-priority interrupt occurs
2. CPU jumps to ISR using vector
3. If higher priority interrupt arrives → nested execution
4. Background tasks handled via software interrupts

Diagram

Interrupt Source → Interrupt Controller → CPU

↓

Priority + Vector Table

Justification

- Vectored interrupts reduce response time
- Nested interrupts ensure critical task execution
- Software interrupts handle non-critical tasks

Conclusion

System achieves **low latency (<10 μs) and efficient multitasking**.

Code of Conduct:

I Monal Raj.B (192511176) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Monal Raj*

MARKS ALLOCATION

	Problem Understanding (25%)	Constraint Analysis (25%)	Solution Development (25%)	Application of Concepts (15%)	Justification (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

- Use vectored interrupts for high-priority devices
- Use software interrupts for background tasks

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MARKS ALLOCATION

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Q1	1.5	1	1	1	1	5.5
Q2	1	1.5	1	.5	1	5
Q3	1.5	2	1.5	.5	.5	6
Q4	1	1.5	1.5	1	.5	5.5
Q5	1	1.5	1	1	1	5.5

27.5

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